AI Weather Wise

REQUIREMENT ANALYSIS

PROJECT OBJECTIVE

Develop a secure, scalable REST API combining real-time weather ingestion and forecasting with Google Gemini AI for personalized outdoor activity recommendations, daily weather summaries, and severe alert insights.

FUNCTIONAL REQUIREMENTS

- User registration and login with JWT generation.
- Protected profile and location preferences access.
- Create, view, update, and delete saved favorite locations.
- Search weather telemetry by city name, coordinates, or date range.
- Fetch real-time weather telemetry and multi-day forecast data.
- Generate AI weather summaries and clothing/activity recommendations.
- Generate AI severe weather alerts and travel advisories.
- Return structured JSON responses and consistent errors.

DATA REQUIREMENTS

User: _id, name, unique email, hashed password, defaultUnitPreference (metric/imperial), timestamps.

Location / Weather Preference: _id, user reference, cityName, countryCode, latitude, longitude, isFavorite, timestamps.

Weather Categories & Metrics: Temperature, humidity (≥ 0%), windSpeed (≥ 0 km/h), uvIndex (≥ 0), precipitationProbability, weatherCondition (Clear, Rain, Snow, Thunderstorm, Clouds, Mist).

SECURITY & NON-FUNCTIONAL REQUIREMENTS

JWT protection for protected routes; bcrypt.js password hashing; Mongoose schema validation; environment variables for secrets; modular MVC structure; centralized error handling; maintainable RESTful design.

SOFTWARE REQUIREMENTS

Windows 10/11, macOS, or Linux; Node.js v16+; npm v8+; Express.js; MongoDB; VS Code; Postman/Thunder Client; openweather-api / axios; express, mongoose, dotenv, bcryptjs, jsonwebtoken, cors, nodemon, @google/generative-ai.

HARDWARE REQUIREMENTS

Intel i5 8th Gen+/Ryzen 5 equivalent; 8 GB RAM minimum, 16 GB recommended; 1 GB available workspace.